

Math Hard Question 1

1

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ABC

In the given expression, a and c are positive constants. If $px + q$ is a factor of the expression,

$$ax^2 + 176x + c = (px + q)(rx + s),$$

where p and q are positive constants, what is the greatest possible value of ac ?

2

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ABC

In the given expression,

$$34z^{18} + bz^9 + 30,$$

b is a positive integer. If $qz^9 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b ?

3

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ABC

The expression

$$6x^4 + 17x^2 + 7$$

can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive non-integers.

What is the value of $a + c$?

4

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ABC

The given quadratic function

$$54x^4 + 219x^2 + 105$$

has factors in the form $(k)(ax^2 + b)(cx^2 + d)$. If a, b, c, d, k are integers, what is the smallest possible value of ab ?

5

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ABC

The equation of a parabola is written in the form

$$y = ax^2 + bx + c,$$

where a, b, c are constants. When graphed in the xy -plane, the parabola has vertex $(-1, 4)$ and does not intersect the x -axis. Which of the following could be the value of $a - b - c$?

A. 2

B. 4

C. -1

D. -6

6

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ABC

A quadratic function with vertex $(1, 32)$ can be written as

$$-ax^2 + bx + c \quad (a, b, c \text{ are all positive integers}).$$

What is the maximum value of b ?

7

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ABC

An object's speed is increasing at a rate of 7.7 meters per second squared. What is this rate in **miles per minute squared**, rounded to the nearest tenth? (Use 1 mile = 1609 meters.)

A. 0.3

B. 17.2

C. 206.5

D. 209

8

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ABC

The function f is defined by

$$f(x) = ab^{\frac{x}{n}},$$

where a, b, n are constants, and b, n are integers. If $f(2) = 6$ and $f(4) = 150$, what is the value of $f(5)$?

9

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ABC

The graph of $y = f(x)$ in the xy -plane has a vertex at $(4, -7)$, contains the point $(3, -9)$, and has a y -intercept at $(0, a)$. The graph of $y = 6 \cdot f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

10

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ABC

The function f is defined by

$$f(x) = ax^2 + bx + c,$$

where a, b, c are constants. The graph of $y = f(x)$ passes through the points $(9, 0)$ and $(-2, 0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

A. 8

B. 7

C. -6

D. -12

11

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ABC

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

A. $3x^2 + 7x + 14b$

B. $3x^2 + 22x + 14b$

C. $3x^2 + 30x + 14b$

D. $3x^2 + 37x + 14b$